QUIZ 1 -MTH301-page1-4

Graded

Student

YATIN PREETAM BHOJWANI

Total Points

3 / 10 pts

Question 1

Q1 — **0** / 5 pts

✔	**+ 0 pts** Completely wrong/not attempted

Question 2

Q2 — **3** / 5 pts

✔	**+ 3 pts** partial correct

ANALYSIS - I : MTH 301 : QUIZ: 2025-26

Name: YATIN PREETAM BHOJWANI Roll No: 241211

- Write your name cleanly in **CAPITAL** letters and roll number inside the boxes.
- Write answers in the space provided only. Maximum marks: 20 Time: 6:45 pm - 8:00 pm.

1. (a) Can we find a metric space (M, d) having exactly 2025 open sets? Justify your answer. [2.5]

   (b) Let $f(x) = \log\left(\frac{23+\sin(2\pi x)^7}{1+e^{2x}}\right)$ and $x_n = \frac{n+\sin n}{n}$ be a sequnece for $n \in \mathbb{N}$. Calculate $\limsup f(x_n)$ and $\liminf f(x_n)$. [2.5]

(a).

(b) $f(n) = \log\left(\frac{23+\sin(2\pi n)^7}{1+e^{2n}}\right)$

$f(x_n) = \log\left(\frac{23+\sin\left(2\pi\left(1+\frac{\sin n}{n}\right)\right)^7}{1+e^{2\sin\left(n+\frac{\sin n}{n}\right)}}\right)$

1	2

2. Let $(\mathbb{R}, d)$ be the usual metric space and $f : \mathbb{R} \to \mathbb{R}$ be continuous. Show that the set $U = \{x \in \mathbb{R} : f(x) > a\}$ for some $a \in \mathbb{R}$ is an open set. Check whether U will be an open set if f is monotone but not continuous. [5]

$(\mathbb{R}, d)$ is the metric space
with $d(x,y) = |x-y|$.
$f: \mathbb{R} \to \mathbb{R}$ is continuous
$U = \{x \in \mathbb{R} : f(x) > a\}$

$f(y) > a \Rightarrow y \in U$ $f(y) \leq a \Rightarrow y \notin U$
~~⇒ the set is open~~ w.l.o.g $f(x)$ is non decreasing
~~the set only has open~~
the fn only has jump

not:

Rough.

set is open if.
for each $x \in A$
$\exists \varepsilon > 0$, s.t. $B_\varepsilon(x) \subseteq A$.
$\boxed{B_\varepsilon x = \{y : \underbrace{|y-x| < \varepsilon}\}}$
for

$x \in \mathbb{R} : f(x) > a$.
consider
$f(x) > a$.
Let ε be δ.
since $f(x)$ is continuous at $x = x_0$.
~~s.t.~~ $|f(x) - f(x_0)| < \varepsilon$.